

ASE6029 Linear Optimal Control Homework #2

- 1) *A system with mixed eigenvalues.* Consider a linear dynamical system $\dot{x} = Ax$ with $x \in \mathbb{R}^n$ where A is diagonalizable and has mixed eigenvalues such that

$$\Re \lambda_1 < 0, \dots, \Re \lambda_s < 0,$$

for some $s < n$ and

$$\Re \lambda_{s+1} \geq 0, \dots, \Re \lambda_n \geq 0.$$

Let v_i and w_i be the (right) eigenvector and the left eigenvector associated with the i -th eigenvalue, λ_i , that is,

$$Av_i = \lambda_i v_i \quad \text{and} \quad w_i^T A = \lambda_i w_i^T.$$

- a) Show that $x(t)$ for an arbitrary initial state $x(0)$ is given by:

$$x(t) = \sum_{i=1}^n e^{\lambda_i t} v_i w_i^T x(0)$$

- b) Show that $x(t) \rightarrow 0$ as $t \rightarrow \infty$ if

$$w_i^T x(0) = 0, \quad \text{for } i = s+1, \dots, n.$$

- c) Show that the above condition is equivalent to the following.

$$x(0) \in \mathbf{span}\{v_1, \dots, v_s\}.$$

In other words, $x(t) \rightarrow 0$ as $t \rightarrow \infty$ in this case.

- 2) *Two easy optimization problem.* Find the solutions to the following optimization problems.

- a) For a given $A \in \mathbb{R}^{n \times n}$, find $x \in \mathbb{R}^n$ that solves:

$$\begin{aligned} & \underset{x}{\text{minimize}} && x^T A x \\ & \text{subject to} && \|x\| \leq 1. \end{aligned}$$

- b) For a given $A \in \mathbb{R}^{m \times n}$, find $x \in \mathbb{R}^n$ that solves:

$$\begin{aligned} & \underset{x}{\text{minimize}} && \|Ax\| \\ & \text{subject to} && \|x\| \leq 1. \end{aligned}$$

- 3) *Solution to linear dynamical systems with inputs and outputs.* Consider the linear time-invariant system:

$$\dot{x}(t) = Ax(t) + Bu(t), \quad x(0) = x_0,$$

where $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, and $u : [0, T] \rightarrow \mathbb{R}^m$ is continuous.

Then the state trajectory, $x(t)$ for $0 \leq t \leq T$, is given by:

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-\tau)} B u(\tau) d\tau.$$

Explain why.